The Church-Rosser property survives transitive closure: if a relation has CR then its transitive closure does too

The Open Logic Project contributors

openlogic:lambda-calculus/church-rosser:local-snapshot-20261013;
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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two unwrapped chains, the align-environment definition of the grid with its interior-cell explanation, and the row/column closure are all verbatim in the span; nothing is delegated.

Editorial note. The label thm:str names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Church-Rosser property for the base relation. The hypothesis of the theorem itself: xredone satisfies CR, invoked at proof lines 656-657 to fill each interior grid cell. Guarantees the term R with N_{i-1,j} xredone R and N_{i,j-1} xredone R for every interior cell.
- Transitive closure as reduction sequences. The unit's notation: xred is the smallest transitive relation containing xredone (introduced in the statement at source lines 622-626). Reads each grid row and column back as a sequence of xred steps in the final two paragraphs.

Theorem 0.1. *If a relation $=_\beta$ satisfies the Church–Rosser property, and $\to$ is the smallest transitive relation containing $=_\beta$, then $\to$ satisfies the Church–Rosser property too.*

Proof. Suppose

$$M =_\beta P_1 =_\beta \cdots =_\beta P_m \text{ and}$$
$$M =_\beta Q_1 =_\beta \cdots =_\beta Q_n.$$

We will prove the theorem by constructing a grid N of terms of height is $m+1$ and width $n+1$. We use $N_{i,j}$ to denote the term in the i-th row and j-th column.

We construct N in such a way that $N_{i,j} =_\beta N_{i+1,j}$ and $N_{i,j} =_\beta N_{i,j+1}$. It is defined as follows:

$$\begin{aligned} N_{0,0} &= M \\ N_{i,0} &= P_i && \text{if } 1 \leq i \leq m \\ N_{0,j} &= Q_j && \text{if } 1 \leq j \leq n \end{aligned}$$

and otherwise:

$$N_{i,j} = R$$

where R is a term such that $N_{i-1,j} =_\beta R$ and $N_{i,j-1} =_\beta R$. By the Church–Rosser property of $=_\beta$, such a term always exists.

Now we have $N_{m,0} =_\beta \cdots =_\beta N_{m,n}$ and $N_{0,n} =_\beta \cdots =_\beta N_{m,n}$. Note $N_{m,0}$ is P and $N_{0,n}$ is Q. By definition of $\to$ the theorem follows. $\square$